networking site for scientists to share papers and queries, reviewed 100 sleep studies in India and concluded that 1 in 4 Indians was afflicted with insomnia and nearly half the Indian population suffered from sleep issues.

But what counts as good sleep? According to National Sleep Foundation guidelines, it is when you can go off to sleep within 30 minutes or so of bidding everyone good night, don't wake up for more than five minutes at night, and if you do, go back to sleep immediately and wake up feeling refreshed in the morning. Adults should sleep at least seven or more hours at night, according to the American Academy of Sleep Medicine (AASM) and Sleep Research Society (SRS).

Any imbalance in this cycle of wakefulness and rest interferes with the body's natural reset mechanism, triggering a range of adverse biological reactions. "Sleep is not just resting. It helps regulate energy storage, works on self-repair (the reason why those who are sick are recommended to get enough sleep), reorganises memories and learned information. If you don't allow your body to do all this, it will physically and mentally weaken over time," explains Dr Suri. Sleep disorders themselves can be of several kinds. Bengaluru-based doctor Satish Ramaiah, a psychiatrist who also specialises in sleep medicine and is currently executive director, psychiatric services, at the mental health hospital chain Sukoon Health, attempts to characterise the common sleep issues he has found among his patients. "The most common sleep disorder," he says, "is insomnia (reduced quality and/or quantity of sleep), which 30-40 per cent of the population experience at some point in life, of which about 10 per cent cases can become chronic." Other than that, about 10 per cent develop parasomnia, or some form of abnormal experience or behaviour during sleep like sleep talking, sleepwalking, nightmares or night terrors. Another 10-odd per cent report a circadian rhythm disorder, which is a mismatch between a person's required physiological schedule and their sleep patterns. Some 2-4 per cent have obstructive sleep



"Sleep is not just resting. It regulates energy storage, helps self-repair, reorganises memories. If you don't allow your body to do this, it will weaken over time"

DR J.C. SURI
Founder, National Sleep Clinic, Safdarjung Hospital

THE SCIENCE OF SLEEP

WHAT IS SLEEP?

❖ Sleep is a natural biological process which allows the body to rest. During this time, the human body regulates energy storage, works on self-repair and reorganises memories and learned information. Sleep is necessary for the proper performance of nearly every bodily function

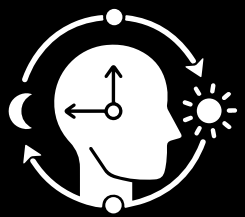
❖ Less sleep or poor quality sleep may lead to a wide range of disorders including hypertension, depression, stroke, heart disorder, stress, cognitive decline, hormonal issues and an increased risk of accidents

WHAT IS GOOD SLEEP?

❖ It is when you can fall asleep within 30 minutes of going to bed, wake up for no more than 5 minutes in the night and, if you do, go back to sleep immediately, and wake up rested in the morning. Adults need a minimum of 7 or more hours of sleep

HOW DOES THE BODY FALL ASLEEP?

❖ Sleep usually follows a circadian rhythm which responds to light cues. When the body senses light is reducing, during night, it starts producing more of the sleep-inducing melatonin hormone, thus helping it to fall asleep



WHAT ARE THE DIFFERENT STAGES OF SLEEP?

❖ When the body sleeps, it goes through cycles where the brain alternates between rest and activity, between different stages of **Non-Rapid Eye Movement (NREM)** sleep and **Rapid Eye Movement (REM)** sleep. The cycle of the three stages of NREM sleep and REM sleep may happen multiple times in a night, with each cycle lasting for 90 to 120 minutes. For an average adult, sleeping for about eight hours gives the body the desired 4-5 rounds of sleep cycle needed to wake up refreshed